

APM02B2 Introduction to Numerical Analysis

Bisection Method

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① Root-finding Methods

② Bisection Method

③ Analysis

Numerical analysis is the study of algorithms for solving problems of continuous mathematics, by which we mean problems involving real or complex variables.

Lloyd N. Trefethen

Root-finding Methods

Problem Statement

Let $I \subset \mathbb{R}$ be an interval of the real numbers and $f : I \rightarrow \mathbb{R}$ be continuous on I .

Generally, we assume that I is a bounded and closed interval. That is,

$$I = [a, b],$$

where $a, b \in \mathbb{R}$ such that $a < b$ (this guarantees that I is not empty).

Problem

Consider the general equation

$$f(x) = 0. \tag{1}$$

We wish to find all $x^* \in I$ that satisfy equation (1). These x^* are called the *roots of the function f* .

Root-finding Methods

Root-finding methods are *iterative methods* to approximate the roots of f , that is, they are methods to approximate the solutions to equation (1).

An *iterative method* is a mathematical procedure that uses an *initial value* x_0 to generate a sequence

$$(x_k)_{k=0}^{\infty} = (x_0, x_1, x_2, \dots)$$

of improving approximate solutions to a problem.

If x^* is a root of f , then ideally we want

$$\lim_{k \rightarrow \infty} x_k = x^*.$$

However, in practice, we shall not let $k \rightarrow \infty$ and only consider a finite number of elements of the sequence.

Iterative Method

Iterative methods are usually defined by a *recurrence relation*

$$x_k = \psi(x_{k-1}) \quad (2)$$

or

$$x_k = \Psi(x_0, x_1, x_2, \dots, x_{k-1}), \quad (3)$$

where ψ is some function describing the functional relationship between x_k and other values in the recurrence and is established by the particular iterative method.

For the recurrence relation (2) we need to specify a single initial condition x_0 , also called the *initial value*.

For the recurrence relation (3) we would need to specify $k - 1$ initial values.

Most methods in numerical analysis are of the form (2).

Bisection Method

Bisection Method

The *bisection method*'s name derives from what it does: “**bisect**” means “to divide something in two, usually equal, parts”.

It is an iterative method for approximating the roots of a general functional equation (1). It is a robust method but converges slowly.

It is also known by the following names: interval halving method, binary search method¹, and dichotomy method.

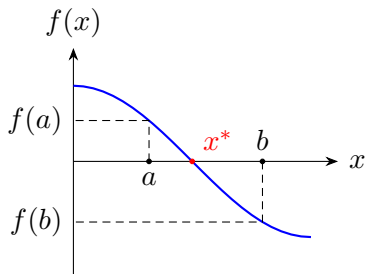
Out of all the root-finding methods we are going to study, it requires the least information about f —it only requires that

- f be continuous on an interval which contains the root, and
- the sign of f at two points on this interval.

¹Not to be confused with binary search from Computer Science.

Bisection Method

Insight



The bisection method relies on a simple insight: if an interval $I = [a, b]$ can be found such that

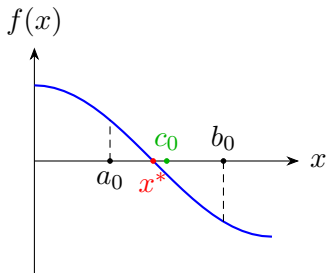
$$f(a)f(b) < 0,$$

and f is continuous on this interval, then f has *at least one real root* x^* on the interval.

This follows from the intermediate value theorem.

Bisection Method

Methodology



Let $I = [a, b] = [a_0, b_0] = I_0$.

Calculate the midpoint of I_0 :

$$c_0 = \frac{a_0 + b_0}{2}.$$

This value c_0 is considered the initial approximation to the root x^* .

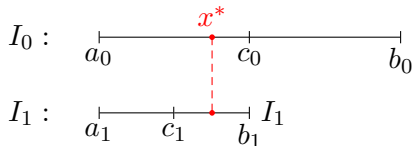
Bisection Method

Methodology

Next, we compare $f(a)f(c_0)$ and $f(c_0)f(b)$:

- If $f(a)f(c_0) < 0$, then $x^* \in [a_0, c_0]$ and we set $I_1 = [a_1, b_1] = [a_0, c_0]$.
- If $f(c_0)f(b) < 0$, then $x^* \in [c_0, b_0]$ and we set $I_1 = [a_1, b_1] = [c_0, b_0]$.

Then we calculate $c_1 = \frac{1}{2}(a_1 + b_1)$.



Repeat the entire process with the newly calculated points until a chosen or specified stopping criterion is met.

Bisection Method

Stopping Criteria

There are three possible stopping criteria² we may use:

$$|c_k - c_{k-1}| < \varepsilon, \quad (4)$$

$$\left| \frac{c_k - c_{k-1}}{c_k} \right| < \varepsilon, \quad (5)$$

$$|f(c_k)| < \varepsilon. \quad (6)$$

where $\varepsilon \in \mathbb{R}$ such that $0 < \varepsilon < 1$ (generally). The symbol ε is referred to as the *tolerance*.

²See slides 40–46 for a longer discussion on errors and stopping criteria.

Bisection Method

Algorithm

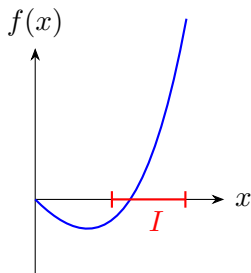
Require: $I = [a, b]$, $f : I \rightarrow \mathbb{R}$ is continuous on I , $N =$ maximum number of iterations.

- 1: Set $k = 0$, $I_0 = [a_0, b_0] = [a, b] = I$.
- 2: **while** $k \leq N$ **do**
- 3: Calculate $c_k = \frac{1}{2}(a_k + b_k)$ and $f(c_k)$.
- 4: **if** $f(c_k) = 0$ **or** $|c_k - c_{k-1}| < \varepsilon$ **then**
- 5: **return** c_k (and stop algorithm)
- 6: **else**
- 7: **if** $f(a_k)f(c_k) < 0$ **then**
- 8: Set $I_{k+1} = [a_{k+1}, b_{k+1}] = [a_k, c_k]$
- 9: **else**
- 10: Set $I_{k+1} = [a_{k+1}, b_{k+1}] = [c_k, b_k]$
- 11: **end if**
- 12: $k = k + 1$
- 13: **end if**
- 14: **end while**

Bisection Method

Example 1 (Adapted from Süli and Mayers 2003)

Consider the function $f(x) = e^x - 2x - 1$ for $x \in [1, 2]$. Use the bisection method to approximate the root x^* in $[1, 2]$ for the given tolerance 10^{-1} .



Consider the figure to the left. Clearly, there is a root of f in the interval $I = [1, 2]$.

We verify this by calculating

$$f(1) \approx -0.2817 < 0, \quad f(2) \approx 2.3891 > 0$$

and noting $f(1)f(2) < 0$.

Let $\varepsilon = 10^{-1} = 0.1$.

Bisection Method

Example (continued)

Let $I_0 = [a_0, b_0] = [1, 2]$. Note that

$$|I_0| = |b_0 - a_0| = 1 > 0.1.$$

Midpoint:

$$c_0 = \frac{a_0 + b_0}{2} = \frac{1 + 2}{2} = \frac{3}{2} = 1.5 \quad \text{with} \quad f\left(\frac{3}{2}\right) \approx 0.4817.$$

Compare:

$$f(1)f\left(\frac{3}{2}\right) < 0, \quad f\left(\frac{3}{2}\right)f(2) > 0$$

Therefore, $x^* \in [a_0, c_0]$.

Bisection Method

Example (continued)

Set $I_1 = [a_1, b_1] = \left[1, \frac{3}{2}\right] = [a_0, c_0]$.

Error:

$$|I_1| = |b_1 - a_1| = \left| \frac{3}{2} - 1 \right| = \frac{1}{2} = 0.5 > 0.1,$$

Midpoint:

$$c_1 = \frac{a_1 + b_1}{2} = \frac{1 + \frac{3}{2}}{2} = \frac{5}{4} = 1.25, \quad f(c_1) = f\left(\frac{5}{4}\right) \approx -9.6570 \times 10^{-3}$$

Compare:

$$f(1)f\left(\frac{5}{4}\right) > 0, \quad f\left(\frac{5}{4}\right)f\left(\frac{3}{2}\right) < 0$$

Therefore, $x^* \in [c_1, b_1]$.

Bisection Method

Example (continued)

Set $I_2 = [a_2, b_2] = \left[\frac{5}{4}, \frac{3}{2}\right] = [c_1, b_1]$

Error:

$$|I_2| = |b_2 - a_2| = \left| \frac{3}{2} - \frac{5}{4} \right| = \frac{1}{4} = 0.25 > 0.1.$$

Midpoint:

$$c_2 = \frac{a_2 + b_2}{2} = \frac{\frac{5}{4} + \frac{3}{2}}{2} = \frac{11}{8} = 1.375 \quad f(c_2) = f\left(\frac{11}{8}\right) \approx 0.2051$$

Compare:

$$f\left(\frac{5}{4}\right) f\left(\frac{11}{8}\right) < 0, \quad f\left(\frac{11}{8}\right) f\left(\frac{3}{2}\right) > 0$$

Therefore, $x^* \in [a_2, c_2]$.

Bisection Method

Example (continued)

Set $I_3 = [a_3, b_3] = \left[\frac{5}{4}, \frac{11}{8}\right] = [a_2, c_2]$

Error:

$$|I_3| = |b_3 - a_3| = \left| \frac{11}{8} - \frac{5}{4} \right| = \frac{1}{8} = 0.125 > 0.1.$$

Midpoint:

$$c_3 = \frac{a_3 + b_3}{2} = \frac{\frac{5}{4} + \frac{11}{8}}{2} = \frac{21}{16} = 1.3125 \quad f(c_3) = f\left(\frac{21}{16}\right) \approx 0.0905$$

Compare:

$$f\left(\frac{5}{4}\right) f\left(\frac{21}{16}\right) < 0, \quad f\left(\frac{21}{16}\right) f\left(\frac{11}{8}\right) > 0$$

Therefore, $x^* \in [a_3, c_3]$.

Bisection Method

Example (continued)

Set $I_4 = [a_4, b_4] = \left[\frac{5}{4}, \frac{21}{16}\right] = [a_3, c_3]$.

Error:

$$|I_4| = |b_4 - a_4| = \left| \frac{21}{16} - \frac{5}{4} \right| = \frac{1}{16} = 0.0625 < 0.1.$$

Midpoint:

$$c_4 = \frac{a_4 + b_4}{2} = \frac{\frac{5}{4} + \frac{21}{16}}{2} = \frac{41}{32} \approx 1.28125, \quad f(c_4) = f\left(\frac{41}{32}\right) \approx 0.0386$$

Bisection Method

Example (continued)

Since $|I_4| = |b_4 - a_4| < \varepsilon$ (the interval length is within the specified tolerance), we stop the bisection method.

The root x^* is approximately $c_4 = 1.28125$, which we also indicate as

$$x^* \approx 1.28125.$$

Bisection Method

Example (continued)

Our calculations in this example are quite verbose, we can abbreviate it into a table:

k	a_k	c_k	b_k	$ c_k - c_{k-1} $	$ f(c_k) $	$ b_k - a_k $
0	1	1.5	2		0.4817	1
1	1	1.25	1.5	0.25	0.0097	0.5
2	1.25	1.375	1.5	0.125	0.2051	0.25
3	1.25	1.3125	1.375	0.0625	0.0905	0.125
4	1.25	1.28125	1.3125	0.03125	0.0386	0.0625

The last three column indicate the possible error measures for the bisection method.

Can you see the difference between the different possible stopping criteria for the bisection method for this example using this table?

Analysis

Bisection Method

Convergence analysis

The question arises whether the sequence $(c_k)_{k=0}^{\infty}$ of approximations generated by the bisection method will actually converge to a root x^* of f .

That is, will

$$c_k \rightarrow x^* \quad \text{as} \quad k \rightarrow \infty?$$

Equivalently, we may ask if

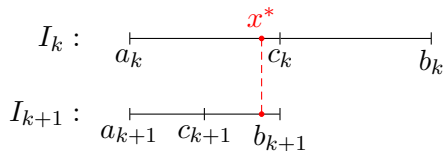
$$|c_k - x^*| \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty$$

or

$$\left| \frac{c_k - x^*}{x^*} \right| \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty?$$

Bisection Method

Convergence analysis



Consider the figure on the left.

Note that

$$c_k - a_k = \frac{b_k - a_k}{2} = b_k - c_k$$

and $|c_k - a_k| = |b_k - c_k|$.

Therefore, we may speak of either subinterval $I_k = [a_k, c_k]$ or $I_k = [c_k, b_k]$ in the following analysis without loss of generality.

Bisection Method

Convergence analysis

Note that we may write

$$c_k = \frac{a_k + b_k}{2} = a_k + \frac{b_k - a_k}{2}.$$

Consider

$$c_k = \frac{a_k + b_k}{2} = a_k + \frac{b_k - a_k}{2}.$$

Subtract x^* on both sides and take the absolute value

$$|c_k - x^*| = \left| a_k + \frac{b_k - a_k}{2} - x^* \right| \leq |a_k - x^*| + \left| \frac{b_k - a_k}{2} \right| \leq \frac{|b_k - a_k|}{2} + \frac{|b_k - a_k|}{2}.$$

Hence,

$$|c_k - x^*| \leq |b_k - a_k|.$$

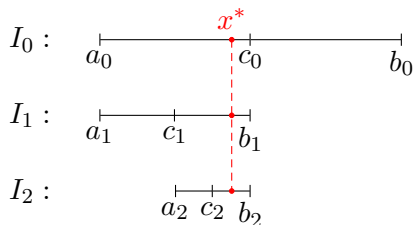
Bisection Method

Convergence analysis

Next, we show that

$$|b_k - a_k| \leq \frac{|b - a|}{2^k}.$$

Note, with the help of the figure to the left, that



$$|I_1| = \frac{|I_0|}{2} \quad \text{and} \quad |I_2| = \frac{|I_1|}{2},$$

where $|I_k|$ represents the length of the k th subinterval. It follows that

$$|I_2| = \frac{|I_1|}{2} = \frac{1}{2} \left(\frac{|I_0|}{2} \right) = \frac{|I_0|}{2^2}.$$

Bisection Method

Convergence analysis

More generally, after k iterations of the bisection method, we have

$$\begin{aligned}|I_k| &= \frac{|I_{k-1}|}{2} \\&= \frac{1}{2} \left(\frac{|I_{k-2}|}{2} \right) = \frac{|I_{k-2}|}{2^2} \\&= \frac{1}{2^2} \left(\frac{|I_{k-3}|}{2} \right) = \frac{|I_{k-3}|}{2^3} \\&\vdots \\&= \frac{1}{2^{k-1}} \left(\frac{|I_0|}{2} \right) = \frac{|I_0|}{2^k}.\end{aligned}$$

Bisection Method

Convergence analysis

Hence,

$$|b_k - a_k| = \frac{|b - a|}{2^k}$$

and

$$|c_k - x^*| \leq \frac{|b - a|}{2^k}.$$

We have proved the following

Theorem 2

Let $[a, b] \subset \mathbb{R}$, where $a < b$, and let $f : [a, b] \rightarrow \mathbb{R}$ be a function such that f is continuous on $[a, b]$ and $f(a)f(b) < 0$. The bisection method generates a sequence $(c_k)_{k=0}^{\infty}$ that approximates a root x^ of f in $[a, b]$ with the property that*

$$|c_k - x^*| \leq \frac{|b - a|}{2^k}, \quad k \in \mathbb{N}.$$

Bisection Method

Convergence analysis

This theorem allows us to calculate the number of iterations to achieve a specified tolerance ε .

Suppose $\frac{|b-a|}{2^k} \leq \varepsilon$. Taking the natural logarithm³ on both sides yields

$$\begin{aligned}\ln\left(\frac{|b-a|}{2^k}\right) &\leq \ln(\varepsilon) &\implies & \ln(|b-a|) - \ln(2^k) \leq \ln(\varepsilon) \\ & &\implies & -k \ln(2) \leq \ln(\varepsilon) - \ln(|b-a|) \\ & &\implies & -k \leq \frac{\ln(\varepsilon) - \ln(|b-a|)}{\ln(2)} \\ & &\implies & k \geq \frac{\ln(|b-a|) - \ln(\varepsilon)}{\ln(2)}\end{aligned}$$

³We could also take the logarithm (base 10).

Bisection Method

Convergence analysis

The expression

$$k \geq \frac{\ln(|b - a|) - \ln(\varepsilon)}{\ln(2)} = \frac{\ln(|b - a|/\varepsilon)}{\ln(2)} \quad (7)$$

informs us the *minimum required number of iterations* that should be performed to achieve the specified tolerance ε .

However, the expression $\ln(|b - a|/\varepsilon)/\ln(2)$ in (7) will usually evaluate to a real number, so we have to choose the next largest natural number.

Note

The inequality (7) requires no information about f itself.

Bisection Method

Convergence analysis

Example 3

Consider the function and interval given in Example 1. Determine the number of iterations required by the bisection method if the given tolerance is $\varepsilon = 10^{-1}$.

We have

$$a_0 = 0, \quad b_0 = 2, \quad \varepsilon = 10^{-1}.$$

Substituting these values into (7), we obtain

$$k \geq \frac{\ln(|1 - 0|) - \ln(10^{-1})}{\ln(2)} \approx 3.3219.$$

We therefore choose $k = 4$.

Bisection Method

Convergence analysis

Example 4

Suppose f is a function with a root in $[0, 2]$ and that $f(0)f(2) < 0$. Determine the number of iterations required by the bisection method if the given tolerance is $\varepsilon = 10^{-5}$.

We have

$$a_0 = 0, \quad b_0 = 2, \quad \varepsilon = 10^{-5}$$

and

$$k \geq \frac{\ln(|2 - 0|) - \ln(10^{-5})}{\ln(2)} \approx 17.60964 \quad \implies \quad k = 18.$$

Bisection Method

Order and rate of convergence

In numerical analysis we are often concerned with *how quickly a numerical method approximates a solution* (or *how quickly a numerical method theoretically converges to a solution if $k \rightarrow \infty$*).

A quantitative measure of convergence is the order and rate of convergence.

Definition 5

The *order of convergence* of an iterative method is the quantity α such that

$$\lim_{k \rightarrow \infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^\alpha} = \mu, \quad (8)$$

where μ is called the *rate of convergence* (or *asymptotic rate of convergence*).

Bisection Method

Order and rate of convergence

Order and rate of convergence is usually determined by *error analysis*⁴ we investigate how errors between subsequent steps of a numerical are related to each other, or how the errors are related to the initial error.

For the bisection method, we have

$$\frac{e_{k+1}}{e_k} = \frac{|c_{k+1} - x^*|}{|c_k - x^*|} = \frac{1}{2}.$$

Therefore, $\alpha = 1$ and $\mu = \frac{1}{2}$. We say that the bisection method converges *linearly*⁵.

⁴See slides 41–46 for definitions of the various errors we encounter in numerical analysis.

⁵Because $\alpha = 1$.

Exercises

Exercises

Questions To Ponder

- What happens with the bisection method when we remove assumptions (e.g., continuity of f , root in $[a, b]$, etc.)?
- What if there are multiple roots of f in the interval $[a, b]$?
- Are there root-finding methods which converge to a root quicker than the bisection method?
- What is the difference between accuracy, precision and tolerance in the context of numerical methods?

Further Reading

-  Burden, Richard L. and Douglas J. Faires (2015). *Numerical Analysis*. 10th. Cengage Learning.
-  Süli, Endre and David F. Mayers (2003). *An Introduction to Numerical Analysis*. Cambridge University Press.

Errors and Stopping Criteria

Stopping Criteria

A *stopping criterion* (plural: *stopping criteria*) is a logical and mathematical test performed to see whether a numerical method is terminated or not.

To facilitate a discussion around this, we introduce the error concepts encountered in the study of applied mathematics and numerical analysis.

First, we introduce two ways in which we can measure error: *absolute error* and *relative error*.

Then we look at the three types of error encountered in computation: *numerical error*, *truncation error*, and *computational error*.

Absolute and Relative Error

Definition 6 (Absolute Error)

The *absolute error* between an actual/solution value x and its approximation $\hat{x}$ is given by

$$e_{\text{abs}} = |x - \hat{x}|. \quad (9)$$

Definition 7 (Relative Error)

The *relative error* between an actual/solution value x and its approximation $\hat{x}$ is given by

$$e_{\text{rel}} = \left| \frac{x - \hat{x}}{x} \right|. \quad (10)$$

Bisection Method

Absolute and Relative Error

Relative error is invariant to scaling (of the values):

$$\left| \frac{(\lambda x) - (\lambda \hat{x})}{\lambda x} \right| = \left| \frac{\lambda(x - \hat{x})}{\lambda x} \right| = \left| \frac{x - \hat{x}}{x} \right|.$$

Relative error expresses the absolute error as a fraction of actual/solution value, and represents absolute error's significance compared to the magnitude of the actual/solution value.

Relative error is closely related to *percentage error*:

$$e_{\text{percentage}} = e_{\text{rel}} \times 100\%.$$

Relative and percentage error are useful when the solution magnitudes vary a lot.

Numerical, Truncation and Computational Error

Numerical error

Let x be our actual/solution value and $\hat{x}$ be our approximation of this value.

Numerical error is the difference between the solutions of our mathematical model (what we have been calling the solution previously) and the solution of the numerical model (what we have been calling the approximations generated by the numerical method previously).

It is quantitatively measured by either the absolute error (9) or relative error (10).

Numerical, Truncation and Computational Error

Truncation error

Most real numbers may be represented as a decimal number with an infinite number of digits after the decimal point. For example,

$$\pi = 3.141592653589793\dots$$

However, computing devices (calculators, cellphones, laptops, desktop personal computers, etc.) only have finite memory to represent numbers. These devices tend to truncate (chop off) or round numbers which are larger than the available memory. For example,

$$\hat{\pi} = 3.14159.$$

This difference between the actual number representation and round-off/truncated number is called the *truncation error* (or *round-off error*).

Numerical, Truncation and Computational Error

Truncation error

Truncation error will also be encountered in numerical methods derived from Taylor series of a function f about a point x_0 :

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

In practice, we use a finite number of terms in the above series to obtain the p th-degree Taylor polynomial:

$$\begin{aligned} T_f(x) &= \sum_{k=0}^p \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \\ &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!} (x - x_0)^2 + \cdots + \frac{f^{(p)}(x_0)}{p!} (x - x_0)^p. \end{aligned}$$

Numerical, Truncation and Computational Error

Computational error

The *computational error* e_c is the sum of the numerical error e_n and truncation error e_r :

$$e_c = e_n + e_r.$$

This represents the total error between any analytical process and our computational and numerical approximation processes.

Stopping Criteria

Difference-based errors and stopping criteria

One of the disadvantages of the absolute error (9) and relative error (10) is that the definitions require the actual solution of the problem—which, in practice, is not known ahead of time or at all.

Most numerical methods use *difference-based errors* as stopping criteria:

$$|x_k - x_{k-1}| < \varepsilon, \quad (11)$$

$$\frac{|x_k - x_{k-1}|}{|x_k|} < \varepsilon. \quad (12)$$

Examples were already encountered in (4) and (5).

Stopping Criteria

Difference-based errors and stopping criteria

What is the difference between $|x_k - x^*|$ and $|x_k - x_{k-1}|$? Is any accuracy lost?

Suppose we have a linearly convergent numerical method (for example, such as the bisection method): $|e_k| \leq q|e_{k-1}|$ (with $0 < q < 1$).

Then

$$\begin{aligned} |x_k - x_{k-1}| &= |x_k - x^* + x^* - x_{k-1}| = |e_k - e_{k-1}| \geq ||e_k| - |e_{k-1}|| \\ &\geq |e_k| - |e_{k-1}| \\ &\geq |e_k| - \frac{1}{q} |e_k| \\ &= \frac{q-1}{q} |e_k| \end{aligned}$$

Stopping Criteria

Difference-based errors and stopping criteria

Hence, we have shown that

$$|e_k| \leq \frac{q}{q-1} |x_k - x_{k-1}|.$$

The absolute error $|e_k| = |x_k - x^*|$ is bounded above by the difference-based error $|x_k - x_{k-1}|$. If $|x_k - x_{k-1}| < \varepsilon$, then we shall have $|e_k| = |x_k - x^*| < \varepsilon$.

Note

This analysis is only applicable to this specific case (linear convergence and $0 < q < 1$) and not in general. It serves to illustrate why we may use difference-based error rather than an error involving the actual solution x^* .